DS-A4: Model Comparison, Calibration, and Capacity-Constrained Review

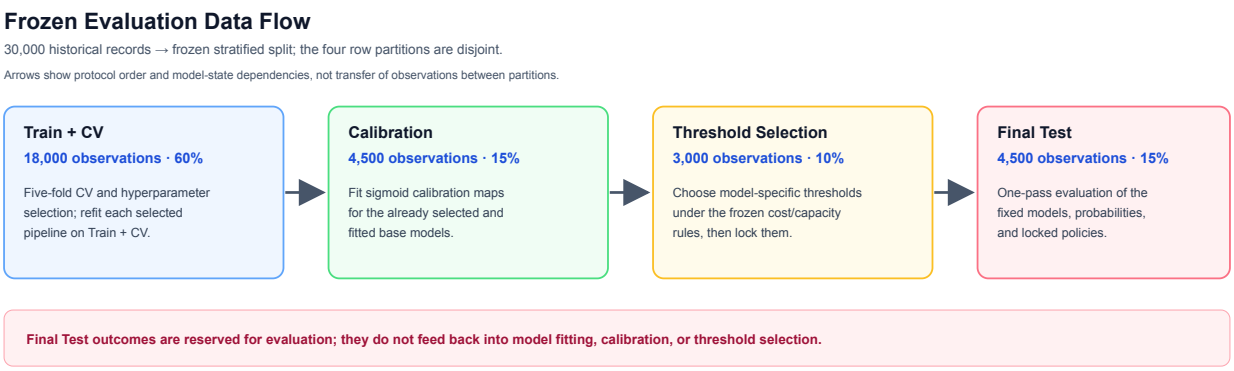
Research question

This study compares Logistic Regression, Random Forest, and HistGradientBoosting for next-month default prediction in the UCI Taiwan credit-card cohort. It separates three questions: how well scores rank positives, how closely probabilities match observed frequencies, and which records a cost- and capacity-constrained rule would select for review. The last is a policy simulation; no intervention outcomes are observed.

AP summarizes precision–recall ranking and is sensitive to positive-class prevalence. ROC-AUC summarizes ranking across thresholds and is theoretically invariant to prevalence when class-conditional score distributions stay fixed. Neither metric chooses an operating threshold. Similar ranking scores can yield different decisions at a cutoff, especially when score calibration differs. Here each model's threshold is selected and locked on the separate Threshold Selection partition.

Cohort and frozen workflow

The 30,000-row historical Taiwan cohort contains a binary next-month outcome and historical predictors. The natural 22.12% positive prevalence is retained. Demographic fields are excluded to focus primary prediction on account history and avoid using those attributes directly; aggregate auditing is supplementary and does not guarantee fairness. Observed PAY codes -2 and 0 remain separate categories because the UCI description does not document their meanings. The dataset describes monthly measures from April through September 2005; end-of-September is an assumed cutoff. No intramonth feature-availability timestamps are supplied, so the availability of all September fields at that cutoff and the absence of temporal look-ahead cannot be fully audited. The random split supports same-cohort evaluation, not future years, economic or regulatory change, contemporary applicants, or other populations.



The four partitions are disjoint. Arrows in the diagram show protocol order and model-state dependencies, not transfer of rows between partitions. Final Test outcomes are reserved for one-pass evaluation and do not feed back into fitting, calibration, or threshold selection.

Paired comparison

All models use the same saved folds and test rows, supporting paired comparisons. Logistic Regression is the interpretable reference; the tree models test nonlinear alternatives. The final bootstrap intervals condition on the selected, fitted models and calibrators.

Frozen CV selection

The primary rule selects the configuration with the highest mean five-fold non-interpolated AP on the shared saved folds. This selects hyperparameters; it is not the final performance estimate. The Final Test remains reserved until model, calibration, and threshold decisions are fixed. All 180 planned fold/configuration records and fit times are retained; equal configuration counts do not imply equal compute time.

	model	C	class_weight	selected_config	mean_cv_AP	sd_cv_AP	max_depth
0	Logistic Regression	3.0	NaN	11	0.542502	0.010896	NaN
1	Random Forest	NaN	NaN	4	0.556518	0.017410	8.0
2	HistGradientBoosting	NaN	NaN	1	0.560583	0.013669	NaN

```
Fold/configuration rows: 180 ; failed folds: 0
Convergence-warning records: 0
```

Final Test discrimination and probability quality

AP and ROC-AUC summarize ranking across thresholds; AP is prevalence-sensitive, while ROC-AUC is theoretically prevalence-invariant when class-conditional score distributions are unchanged. Neither identifies an operating threshold. Brier score and frozen ten-bin ECE summarize probability quality. The cohort's natural prevalence was retained. Applying one numerical probability threshold to differently calibrated scores can select different records; this analysis instead selects a model-specific threshold on separate data. Final Test metrics are reported after those choices were locked.

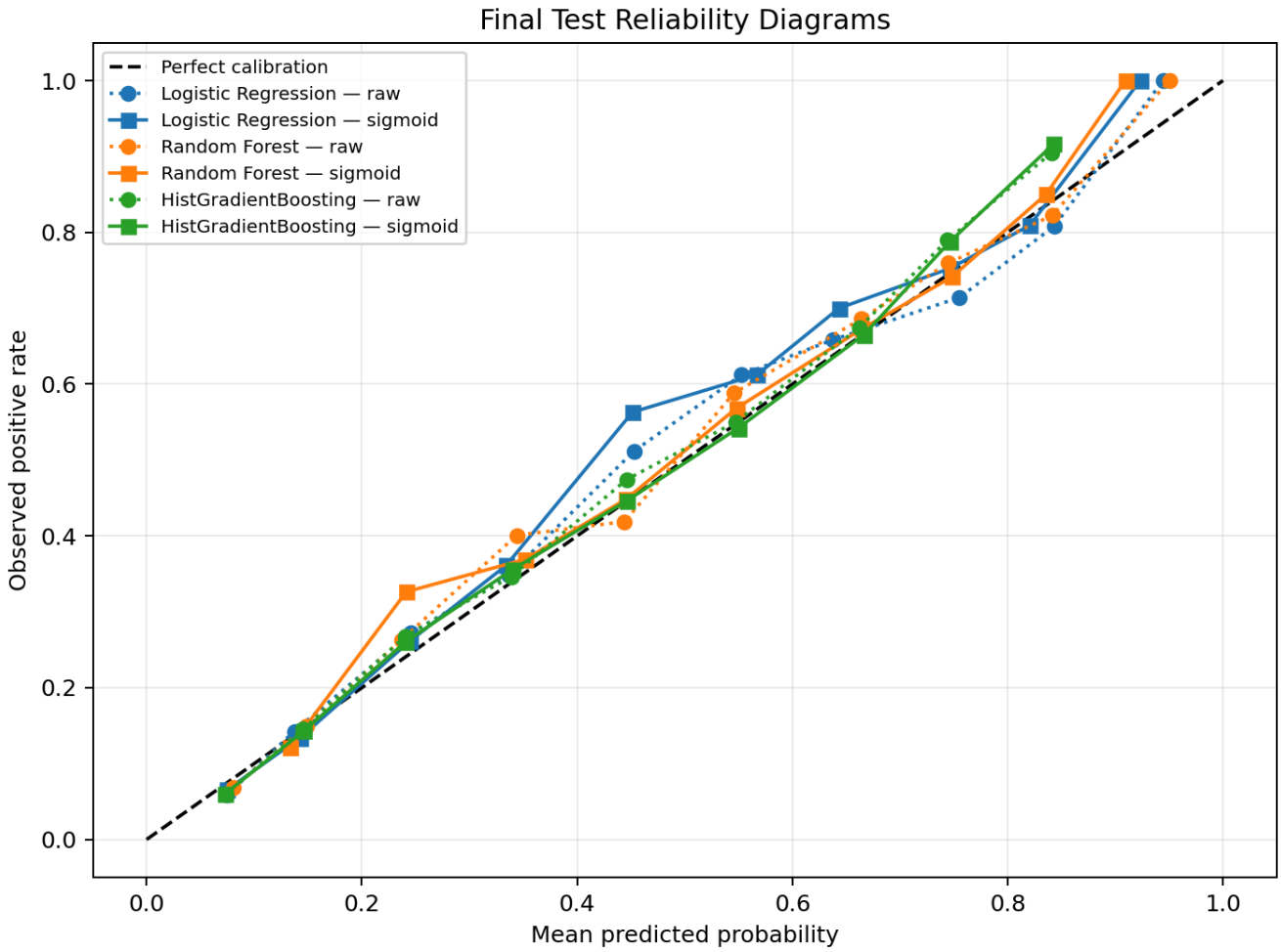
Model	AP	ROC-AUC	Brier (raw)	Brier (sigmoid)	ECE (raw)	ECE (sigmoid)
Logistic Regression	0.5574	0.7694	0.1353	0.1355	0.0137	0.0154
Random Forest	0.5750	0.7820	0.1327	0.1337	0.0132	0.0178
HistGradientBoosting	0.5798	0.7882	0.1320	0.1319	0.0111	0.0106

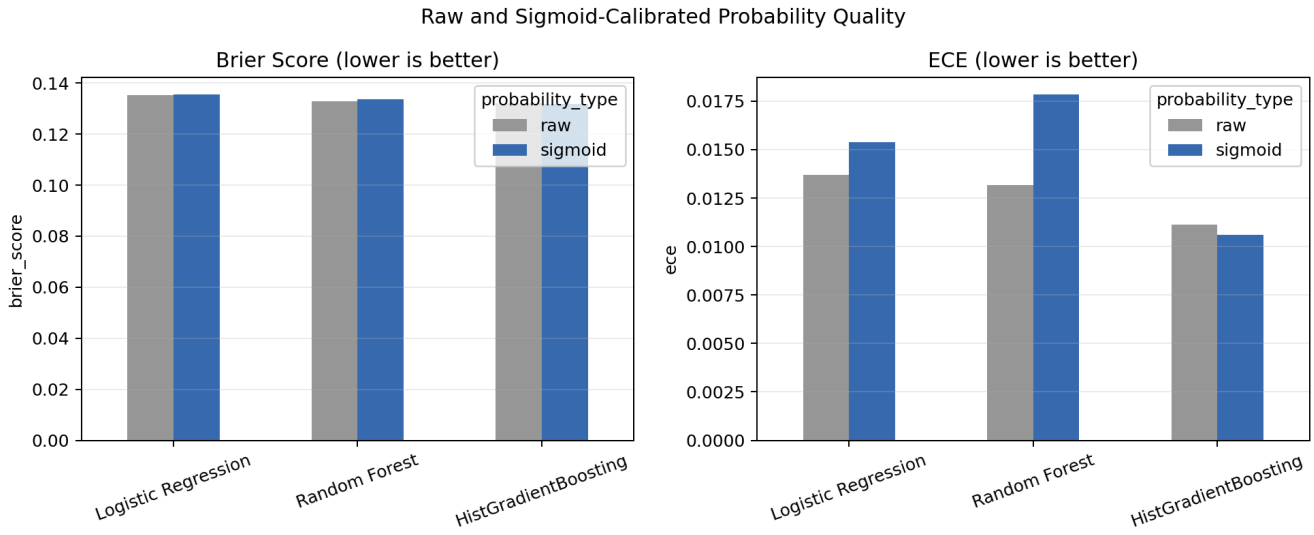
Calibration diagnostics

On Final Test, sigmoid calibration worsened LR Brier/ECE from 0.13528/0.01370 to 0.13553/0.01538, and RF from 0.13274/0.01315 to 0.13368/0.01785. HGB improved slightly, from 0.13200/0.01113 to 0.13192/0.01060. AP and ROC-AUC were unchanged for each model, consistent with the fitted monotonic mapping preserving within-model order, subject to numerical ties.

Saved LR reliability bins show descriptive redistribution. Bin 1 grew from 2,609 to 2,778 records; its weighted absolute-gap contribution to ECE rose from about 0.00319 to 0.00531. Bin 4 grew from 86 to 103 records and its contribution rose from about 0.00113 to 0.00256. Other bins offset part of the increase. This decomposition explains how the fixed-bin ECE changed, but bin movement does not identify why scores moved.

Possible explanations include differences in original score distributions and how closely a sigmoid mapping fits each model's calibration errors. LR's probabilistic form does not guarantee calibration; tree probability averaging does not establish a particular calibration pattern. These mechanisms were not tested. Calibration is empirical here, not a guaranteed model-family benefit.





Paired uncertainty and multiplicity

The saved table contains 24 model × score-type × metric point estimates and 24 model-specific 95% percentile intervals. Paired contrasts follow the protocol: RF vs LR and HGB vs LR across two score types and four metrics, or 16 non-baseline comparisons; the eight LR-vs-itself rows are zero reference values. These are marginal intervals, not a set of 24 independent hypothesis tests, and no family-wise adjustment was applied.

The paired bootstrap uses common test-row draws and conditions on frozen hyperparameters, fitted models, calibrators, and predictions. It excludes selection/refit variability and future temporal or population change. With 2,000 replicates, endpoints have finite Monte Carlo precision; fixed-bin ECE is non-smooth, which can affect percentile interval accuracy. These qualifications do not invalidate the prespecified analysis.

Comparison	Metric	Difference	95% CI lower	95% CI upper
Random Forest vs Logistic Regression	average_precision	0.0176	0.0027	0.0316
HistGradientBoosting vs Logistic Regression	average_precision	0.0224	0.0108	0.0344
Random Forest vs Logistic Regression	roc_auc	0.0126	0.0020	0.0222
HistGradientBoosting vs Logistic Regression	roc_auc	0.0188	0.0093	0.0281

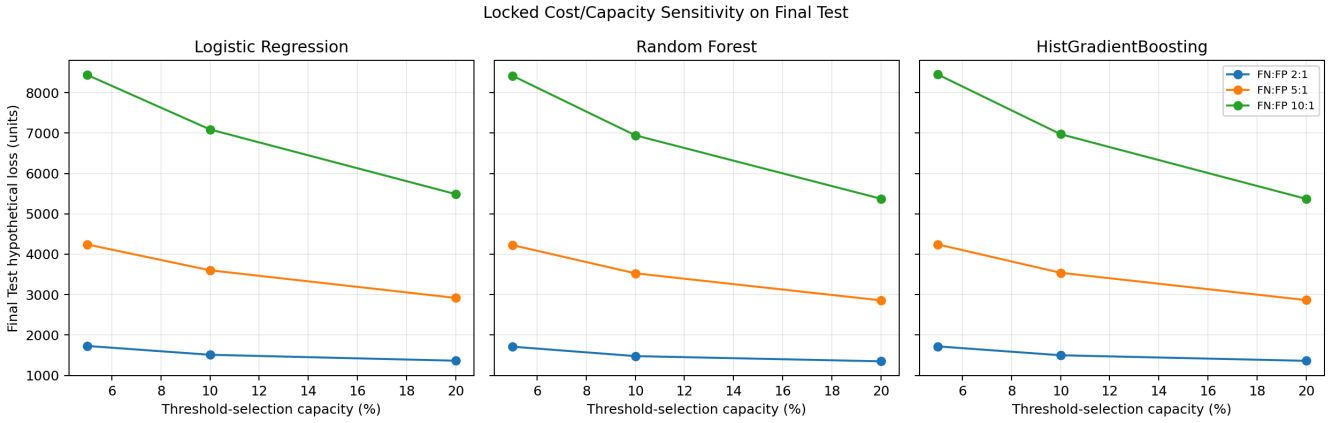
Threshold decisions and sensitivity

The primary policy minimizes `FP + 5 × FN` on Threshold Selection data, subject to at most 10% flagged there; each calibrated threshold is then locked and applied to Final Test. Test flag rates are

observed rates, not future capacity guarantees. FP/FN costs are hypothetical units and do not measure intervention effects.

HGB has the highest observed AP/AUC, while RF has the lowest primary test loss: 3,528 versus 3,541 for HGB, a 13-unit point difference. No direct paired uncertainty interval for this fixed-policy loss difference was reported. Across all nine prespecified sensitivity scenarios, RF has the lowest test loss in eight and HGB in one; the figure summarizes the full grid. Top-K is a separate fixed-count policy and may select different records from a probability threshold.

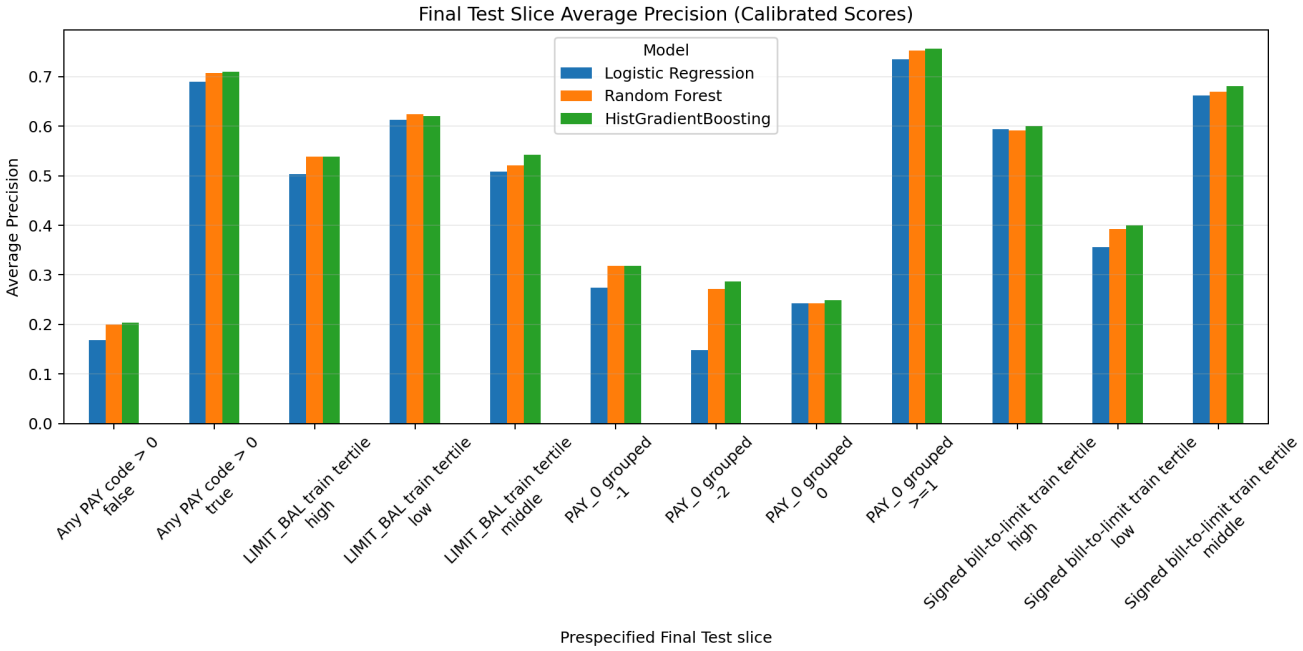
Model	Threshold	TP	FP	FN	Flags	Hypothetical test loss
LR	0.5672	298	118	697	416 (9.2%)	3603
RF	0.5714	312	113	683	425 (9.4%)	3528
HGB	0.5915	309	111	686	420 (9.3%)	3541

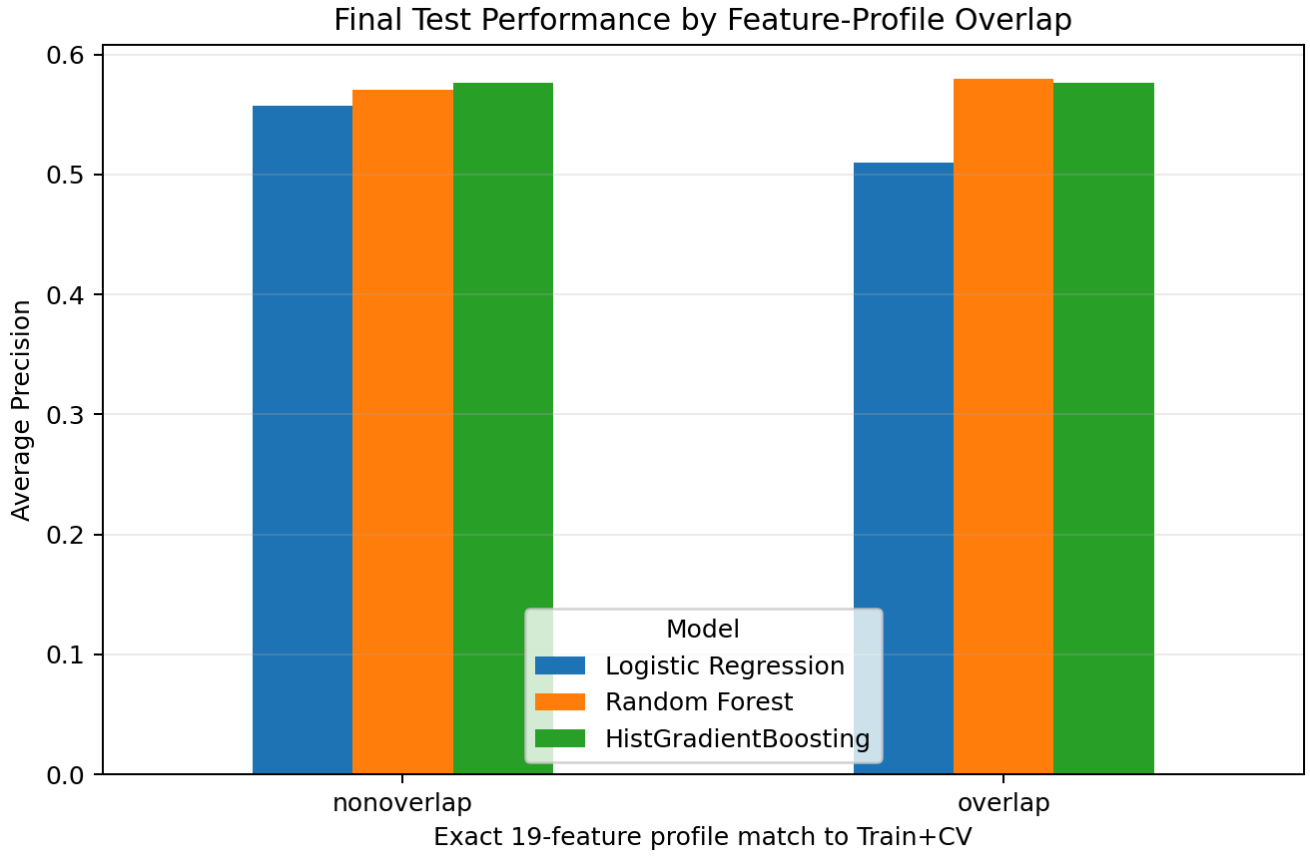


Slices and validity boundaries

In `PAY_0 == 0`, 2,229 Final Test rows contain 285 positives; LR/RF/HGB threshold-selected TPs are 0/1/0. In the all-six-PAY-states-non-positive slice, 3,042 rows contain 357 positives and TP counts are 0/0/0. The definitions overlap but are not equivalent: a row with `PAY_0 = 0` can have positive values in other PAY fields. The all-six slice has AP 0.1678–0.2035 against 11.7% prevalence, so zero selected TPs do not imply zero ranking information. These slice results are descriptive and have no dedicated intervals.

Exact 19-feature profiles overlap between Train+CV and 127 Test rows. This may reduce profile-level split independence but does not prove repeated identities. UCI describes monthly data for April–September 2005, but no intramonth timestamps verify feature availability at the assumed end-of-September prediction cutoff. Potential look-ahead cannot be fully audited; the evidence does not establish that it occurred. A fixed random split does not test future years, economic/regulatory change, contemporary applicants, or other populations. The label also does not identify the effect of intervention.


Final Test Slice Average Precision (Calibrated Scores)
Model
Logistic Regression
Random Forest
HistGradientBoosting
Average Precision
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Any PAY code > 0 false
Any PAY code > 0 true
LIMIT_BAL train tertile high
LIMIT_BAL train tertile low
LIMIT_BAL train tertile middle
PAY_0 grouped -1
PAY_0 grouped -2
PAY_0 grouped 0
PAY_0 grouped >=1
Signed bill-to-limit train tertile high
Signed bill-to-limit train tertile low
Signed bill-to-limit train tertile middle
Prespecified Final Test slice


Final Test Performance by Feature-Profile Overlap
Average Precision
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Model
Logistic Regression
Random Forest
HistGradientBoosting
nonoverlap
overlap
Exact 19-feature profile match to Train+CV

Conditional conclusion

For further offline, ranking-focused evaluation, HGB is the leading candidate: AP is 0.5798 and ROC-AUC is 0.7882, the highest observed values. Its paired AP/AUC differences versus LR have 95% intervals above zero. The saved summaries do not directly compare HGB with RF. Under the frozen primary policy, RF has the lowest observed test loss, 13 hypothetical units below HGB, but no uncertainty interval for that difference was reported. LR remains the interpretable reference.

These results support a bounded model-comparison conclusion, not deployment. A future recommendation would require temporally representative and external evaluation, verified point-in-time feature availability, a prespecified direct HGB–RF comparison, and evidence on real intervention effects and costs. The student has reviewed and accepted the revised narrative and conditional conclusions. This qualitative review does not imply manual recalculation of every numerical result or independent verification of all predictions.

Artifact map and review status

- `results/cv/` : planned CV records and selected parameters.
- `results/models/` : fitted base models and sigmoid calibrators.
- `results/predictions/` : calibration, threshold-selection, and Final Test predictions.
- `results/statistics/` : metrics, paired bootstrap draws/replicates/intervals, and curve coordinates.
- `results/calibration/` : reliability-bin and calibration-subset evidence.
- `results/slices/` , `results/diagnostics/` , and `results/decisions/` : error slices, profile overlap, and locked policies.
- `results/verification/` : independent numerical checks.
- `../docs/frozen_data_flow_v1.0.svg` : canonical frozen-partition and analysis-sequence diagram.

The default notebook mode validates and presents saved results without retraining. The student has manually verified one RF observation and has reviewed and accepted the revised scientific narrative and conditional conclusions. The final exported PDF and submission summary are prepared separately for student review.

Reproduction and provenance

The data source is the UCI Default of Credit Card Clients dataset, ID 350 (Yeh, 2009; DOI 10.24432/C55S3H), licensed CC BY 4.0. The original archive is intentionally excluded from Git. Its official retrieval URL, expected SHA-256, and integrity-check commands are documented in `data/raw/README.md` ; source and acquisition evidence are in `docs/dataset_audit.md` .

The measured environment is Python 3.10.18 with pinned dependencies in `requirements.txt` . The frozen split and CV seed is 42; the paired-bootstrap seed is 2026. README documents environment setup, the full 180-fit experiment, saved-results inspection in an isolated copy, and independent verification. `scripts/verify_phase2_independently.py` checks the preserved predictions, thresholds, metrics, bootstrap results, and critical slices without fitting models. The AI-use and review scope is recorded in `docs/ai_use_log.md` .